Topic 14: Sub-Workflows, Debugging & Production Readiness

n8n Hands-On Intermediate Assessment - Level 2

1. Document Control

Field	Value
Document Title	Topic 14: Sub-Workflows, Debugging & Production Readiness Requirement Document
Version	1.0 (Approved)
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Prepared By	Lead Automation Architect (Antigravity AI / Tayana Academy)
Target Audience	Automation Engineers, Microservice Architects, DevOps Specialists

2. Executive Summary & Objective

Build a modular master order orchestration pipeline in n8n that delegates domain logic to specialized sub-workflows (Tax Engine and Payment Gateway) using the Execute Workflow node. Enforce strict input contract validation, demonstrate execution tracing and node pinning for debugging, and provide a production-ready Docker Compose environment blueprint.

3. Architecture Diagram

```
[ Manual_Trigger ]
        |
        v
[ Code_PrepareOrderPayload ] (Order: State NY, Subtotal $1,200.00)
        |
        v
[ Execute_TaxSubWorkflow ] ──> Sub-Workflow 1: Tax Engine
        |                       (NY 8.875% -> Tax $106.50, Grand Total $1,306.50)
        v
[ Code_MergeTaxTotals ] (Combines order attributes with tax return)
        |
        v
[ Execute_PaymentSubWorkflow ] ──> Sub-Workflow 2: Payment Gateway
        |                           (Charges $1,306.50 -> Confirmed: ch_sub_998102)
```

```
        v
[ Code_BuildOrchestratorSummary ] (Master Telemetry Audit Receipt)
```

4. Node Configuration Blueprint

Node Name	Type	Operation / Setting	Output Outcome
Manual_Trigger	Manual Trigger	Trigger execution on demand	Execution pulse
Code_PrepareOrderPayload	Code	Emits staged order payload	Master order object
Execute_TaxSubWorkflow	Execute Workflow	Target: Sub-Workflow 1 (Tax Engine)	Computed state tax & totals
Code_MergeTaxTotals	Code	Merges parent order with tax return	Enriched order with grand total
Execute_PaymentSubWorkflow	Execute Workflow	Target: Sub-Workflow 2 (Payment Gateway)	Authorized charge receipt
Code_BuildOrchestratorSummary	Code	Master orchestration telemetry receipt	Confirmed fulfillment audit

5. Test Matrix & Verification Scenarios

Scenario	Order Details	Microservice Operations	Expected Outcome
Master Orchestrator Execution	ORD-MODULAR-901, State: NY, Subtotal: $1,200.00	Tax Engine (8.875%) + Payment Gateway ($1,306.50)	Order fulfilled with zero data loss; auth code ch_sub_998102 issued.
Contract Guardrail Test	Empty payload {} sent to Tax Sub-Workflow	Input contract validation	Throws CONTRACT_VIOLATION: Missing required parameter 'state'.